INCH- POUND

MIL- DTL- 12189J
31 May 2018

SUPERSEDING
MIL- C- 12189H
30 June 1988

DETAIL SPECIFICATION

CLOTH, COATED: BUTYL COATED, TOXICOLOGICAL AGENTS PROTECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of butyl coated nylon cloth.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army Edgewood Chemical Biological Center, ATTN: RDCB- DEL- S, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010- 5424 or emailed to usarmy.APG.ecbc.list.specs- team@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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FEDERAL STANDARDS

FED- STD- 191 - Textile Test Methods
FED- STD- 601 - Rubber: Sampling and Testing

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL- DTL- 12468 - Decontaminating Agent, STB
MIL- C- 19377 - Cloth, Twill, Nylon

DEPARTMENT OF DEFENSE STANDARDS

MIL- STD- 282 - Filter Units, Protective Clothing, Gas- Mask Components and Related Products: Performance Test Methods
MIL- STD- 1487 - Glossary of Cloth Coating Imperfections

2.3 Non- Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM- D618 - Standard Test Method for Conditioning Plastics for Testing
ASTM- D1424- Standard Test Method for Tearing Strength of Fabrics by Falling- Pendulum (Elmendorf- Type) Apparatus

(Copies of this document are available from www.astm.org)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE- AMS- STD- 595 - (R) Colors Used in Government Procurement

(Copies of this document are available from https://www.sae.org/)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Material.

3.2.1 Base cloth. The base cloth shall be cloth nylon, twill conforming to MIL- C- 19377 except that the cloth shall be undyed, and the requirement for bias and bow shall not apply.

3.2.2 Coating compound. The coating compound used for coating the base cloth shall be a composition of butyl rubber formulated to meet the requirements of this document. No natural rubber or synthetic rubber other than butyl, or reclaimed rubber of any kind shall be used (see 4.2.3).

3.2.3 Dusting powder. The dusting powder for application to the coated cloth shall be whiting, talc, or other finely divided mineral material that does not support mildew growth (see 4.2.3). Talc shall not contain asbestos and whiting should contain less than 1 percent quartz.

3.3 Coated cloth (see 6.5). The coating compound specified in 3.2.2 shall be applied to both sides of the base cloth, after which the coated cloth shall be fully vulcanized (see 3.2.2). The coating shall be uniformly distributed over both cloth surfaces, with a minimum of 65 percent to a maximum of 75 percent on one side and the balance of the coating on the other side. The coated cloth shall then be dusted on both sides with a minimum amount of the dusting powder specified in 3.4.3 necessary to prevent blocking. The application of dusting powder shall be controlled during processing or reduced by an after- processing brush or wash, so that both sides of the finished coated cloth shall show no less than “good” resistance to crocking then tested as specified in 4.3.

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3.3.1 Physical requirements. The physical properties of the coated cloth shall conform to the requirements of table I when tested as specified in 4.2.2.

TABLE I. Physical requirements.

Characteristic	Requirement	
	Minimum	Maximum
Weight, overall, oz/sq yd	11.0	13.5
Breaking strength, pounds:		
Initial:		
Warp	180	- -
Filling	170	- -
After accelerated weathering:		
Warp	170	- -
Filling	160	- -
Tearing strength, grams:		
Warp	850	- -
Filling	850	- -
Stiffness, centimeters (warp only) at 700 20F	- -	9.5
Adhesion of coating, lb/2- inch width	8.0	- -
Adhesion of coating after water immersion, pounds	4.0	- -
Hydrostatic resistance, psi:		
Initial	200	--
After strength of coating	200	--
After cold crack	200	--
Blocking, scale rating	--	No, 2
Abrasion resistance	1/	--
Colorfastness to weathering	2/	--

1/ After having been abraded the required number of cycles, no loose or exposed fibers or filaments of the base cloth shall be visible in the abraded portion of the specimen.

2/ The material shall show no “appreciable change in color” when compared to the unexposed material.

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3.3.2 State of cure. The finished coated cloth shall be tightly and uniformly cured in such a manner that there is no evidence of the coating becoming soft and tacky or stiff and brittle when subjected to the solvent resistance test specified in 4.3.3.2.

3.3.3 Requirements for resistance to toxicological agents. The coated cloth shall conform to the requirements in table II when tested as specified in 4.3.3 and 4.3.3.3.

TABLE II. Resistance to toxicological agents requirements.

Characteristic	Requirement
Resistance to toxicological agents, minutes (minimum):	
Initial:	
Mustard (H)	100
GB	200
After accelerated weathering:	
Mustard (H)	75
GB	150
After decontamination:	
Mustard (H)	75
GB	150
Colorfastness after decontamination	1/
Breaking strength after decontamination, pounds of force minimum):	
Warp	170
Filling	160

1/ The material shall show no “appreciable change in color” when compared to the unexposed material.

3.4 Color. The color of the coated cloth (both sides) shall be Olive Drab 177 (color chip # 34088 per SAE AMS- STD- 595).

3.4.1 Matching. Before matching for color, the dusting powder shall be removed from the specimens by wiping with a damp cloth and the specimen shall allowed to air dry. The color shall meet 3.4 when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles. The color shall be a good match to 3.4 under incandescent lamplight at 2300 ± 200 K.

3.5 Width. The minimum overall width, after selvage trimmings, shall be as specified (see 6.2). Selvages shall be trimmed to give a straight, uniform edge.

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3.6 Length and put- up. Each roll shall contain no less than 180 yards not more than 220 yards. The maximum number of pieces per roll shall be four, and no single piece shall be less than 20 yards. The ends of the pieces shall be overlapped and not joined by a seam.

3.7 Odor. The coated cloth shall be free from any objectionable odor. Characteristic odor of coating is not considered objectionable.

3.8 Marking. The words HEAVY SIDE shall be stencilled in legible 1/2- inch black letters on the heavier coated side of each roll. The words shall be located on one edge at the beginning and end of each roll.

3.9 Workmanship. The finished cloth shall conform to the quality of product established by the document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2)
(b) Conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Sample. The first article sample shall consist of 7 yards manufactured using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.2.2 Inspection procedure. The first article sample items shall be examined in accordance with table V and subjected to the inspection procedures specified in table III, VI and 4.3.3.2.

TABLE III. First article inspection procedures.

Characteristic	Requirement Para.	Test method
Initial:		
Mustard (H)	3.3.3	4.4.2
GB	3.3.3	4.4.3
After accelerated weathering:		
Mustard (H)	3.3.3	4.4.4
GB	3.3.3	4.4.5
After decontamination:		
Mustard (H)	3.3.3	4.4.6
GB	3.3.3	4.4.7

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Colorfastness: After decontamination	3.3.3	4.4.8
Breaking strength: After decontamination	3.3.3	4.4.9 and 5100 1/

1/ Method of testing as specified in FED- STD- 191

4.2.3 Certification. Components, materials, and processes listed below shall be accepted on the basis of a contractor's certificate of compliance with the indicated requirements:

a. The coating compound shall be a composition of butyl rubber, and and no natural rubber, synthetic rubber other than butyl, or reclaimed rubber of any kind shall be used.

b. The distribution of coating requirement shall be met.

c. The dusting powder used shall not support mildew growth.

d. The cloth shall be fully vulcanized after coating is applied to both sides.

4.2.4 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.3 Conformance inspection.

4.3.1 Lotting. A lot shall consist of the items produced by one manufacturer, at one plant, from the same materials, under the same manufacturing conditions, and at the same time. Any break in continuous production of 24 hours or more of coated fabric shall be cause for identification of a new lot of the end item. A single lot of coated fabric shall not contain material from more than one lot of base fabric. The lot size shall be expressed in units of linear yards.

4.3.2 End item examination.

4.3.2.1 For yard- by- yard examination. The coated cloth shall be examined for the defects listed in table V. The required yardage of each roll shall be examined on one side only, however, the side shall be alternated so that alternate rolls are examined on the face and back, respectively. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The lot size shall be expressed in units

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of yards. The sample unit shall be 1 linear yard. The number of rolls from which the sample yardage is to be selected shall be as specified in table IV. The sample yardage shall be apportioned equally among the selected rolls.

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TABLE IV. Sample size.

Lot size (yards)	Sample size (rolls)
1,201 to 3,200 inclusive 1/	3
1,201 to 3,200 inclusive	5
3,201 to 10,000 inclusive	8
10,001 to 35,000 inclusive	13
35,001 to 150,000 inclusive	20
150,001 and over	32

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

TABLE V. Yard by yard characteristics.

Category	Characteristics	Sampling and acceptance criteria	Inspection method
Major			
101	No cuts, holes or tears	TABLE IV, Acc 0, Rej 1	VI
102	No abrasion mark or scratch that extends through the coating to a point where base cloth yarns or cloth weave can be seen	TABLE IV, Acc 0, Rej 1	VI
103	No pinhole 1/	TABLE IV, Acc 0, Rej 1	VI
104	No uncoated- area	TABLE IV, Acc 0, Rej 1	VI
105	No pit, blister, tunnel, or delamination of coating	TABLE IV, Acc 0, Rej 1	VI
106	No lump or embedded foreign matter that protrudes so as to be readily abraded from material or easily removed	TABLE IV Acc 0, Rej 1	VI
107	No crease or wrinkle resulting in adhesion of surface or delamination of coating when corrected by manual pressured	TABLE IV Acc 0, Rej 1	VI
108	No spot, stain, or streak, more than 1 inch in combined directions that cannot be readily removed with naphtha or water 2/	TABLE IV Acc 0, Rej 1	VI
109	No spliced- cloth (base cloth or coated) or impression in coating from splicing threads	TABLE IV Acc 0, Rej 1	VI

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TABLE V. Yard by yard characteristics - CONTINUED

Major			
110	Width minimum specified	TABLE IV Acc 0, Rej 1	VI
111	Coating properly applied (i.e. not visibly heavier on one side)	TABLE IV Acc 0, Rej 1	VI
112	Even coating (thin areas where coating is missing or insufficient).	TABLE IV Acc 0, Rej 1	VI
113	No signs of over vulcanization (dry, brittle or burned areas)	TABLE IV Acc 0, Rej 1	VI
114	Dusting powder present	TABLE IV Acc 0, Rej 1	VI
115	No tackiness or soft- coating (coating will adhere and not unroll readily)	TABLE IV Acc 0, Rej 1	VI
Minor			
201	Marking present, legible, on heavier coated side of fabric	TABLE IV Acc 0, Rej 1	VI
202	Color and shade as specified	TABLE IV Acc 0, Rej 1	VI
203	No ripples, waviness or dimensional distor- tion	TABLE IV Acc 0, Rej 1	VI
204	No objectionable odor 3/	TABLE IV Acc 0, Rej 1	VI

1/ The examination for pinholes shall be performed in accordance with the through- light inspection procedure in MIL- STD- 1487.
2/ Clearly visible at normal inspection distance (approximately 3 feet).
3/ Odors of chemicals commonly used in coating compounds shall not be regarded as objectionable.

4.3.2.2 Length examination. Each roll of cloth used in the yard- by- yard examination shall be examined for the characteristics listed below. If the sample rolls does not meet the following criteria the lot shall be rejected.

Total length of roll not less than 180 yards or not more more than 220 yards.
Total length of roll shall be within 2 yards or less than that marked on ticket.
Roll shall contain four pieces or less.
Any piece in roll shall be greater than 20 yards in length.
Ends of pieces in roll shall be overlapped and joined by seam.

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4.3.3 End item Testing. Test results obtained under testing conditions defined in FED- STD- 191, ASTM D618, or FED- STD- 601 will be acceptable except in case of dispute. In dispute cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED- STD- 191.

4.3.3.1 Physical testing. The methods of testing specified in FED- STD- 191, wherever applicable, and as listed in table VI shall be followed. The physical values specified in section 3 apply to the average of determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit for testing shall be four continuous yards full width of the coated cloth. The lot size shall be expressed in units of linear yards, and the sample size (number of sample units) shall be as specified below. The lot shall be unacceptable if any sample unit fails to meet any specified requirement. All test reports shall contain the individual values used in expressing the final results.

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

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TABLE VI. Physical tests

Characteristic	Requirement para.	Test method
Resistance to crocking	3.3	5651 1/
Weight, overall	3.3.1	5041
Breaking strength:	3.3.1	
Initial	3.3.1	5100 8/
After accelerated weathering	3.3.1	4.4.1 and 5100 8/
Tearing strength	3.3.1	ASTM D1424
Stiffness (warp only): At 700 ± 20F	3.3.1	5204 2/
Adhesion of coating	3.3.1	5970
Adhesion of coating after water immersion	3.3.1	5970 7/
Hydrostatic resistance:	3.3.1	
Initial	3.3.1	5512 3/
After strength of coating	3.3.1	5972 and 5512 3/
After cold crack	3.3.1	58711 4/ and 5512 3/
Blocking	3.3.1	5872
Abrasion resistance	3.3.1	5302 5/
Colorfastness to weathering	3.3.1	5671 6/

1/ A dry crocking resistance test only shall be conducted on the coated cloth. Rating shall be performed in accordance with method B, using blue crock cloth.

2/ Specimens for stiffness test at 70° F shall be conditioned at the test conditions for not less than 4 hours prior to the test. All tests shall be made in still air.

3/ The lightly coated side of the cloth shall be exposed to the water pressure.

4/ Except that Method 5512 shall be used for determination of hydrostatic resistance, and that the specimens shall be exposed at a temperature of minus 40° ± 2° F for a period of not less than 4 hours.

5/ The following exceptions to Method 5302 shall apply:

a. The heavily coated side of the cloth shall be tested.
b. No metallic pin shall be used.
c. The specimens shall be abraded 100 multidirectional cycles using fine emery cloth as the abradant.
d. Before cutting the test specimens, the coated cloth shall be cleaned by washing with mild soap and water, rinsing in distilled water, and drying at room temperature.

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6/ The heavily coated side of the specimen shall be exposed and the time of exposure shall be 100 hours.

7/ The coated cloth strips shall be immersed for 2 hours in boiling water, removed, and immersed for 15 minutes in distilled water at 750 $\pm$ 5 ^{o}F. The strips shall then be removed from the distilled water, blotted dry, and allowed to air dry under ambient conditions for 24 hours. The coated cloth strips shall then be tested tor adhesion of coating except that the lightly coated side of the cloth shall be tested.

8/ Specimens shall be prepared in accordance with the coated cloth requirements of method 5100.

4.3.3.2 State of cure testing. This test shall be performed on every separate piece of cloth in every roll. Specimens approximately l inch in diameter shall be taken from the center of the width of the piece at the middle. Each specimen shall be separately tested by submerging in 10 mL of toluene in a 3 to 4- inch diameter water glass. The toluene shall be allowed to contact both sides of the specimen for 15 minutes. The toluene shall then be poured off, and the specimen allowed to dry for 5 minutes at standard test conditions before examination. The piece shall be unacceptable if one or more of the specimens from the piece becomes sort and tacky or stiff and brittle.

4.3.3.3 Toxicological agents resistance testing. Every roll of coated cloth shall have a sample of 3 continuous yards full width removed for testing in accordance with table VII. Any test failure shall be cause for rejection of the roll from which the failed sample was taken.

TABLE VII. End item toxicological agents resistance tests see (6.4)

Characteristic	Requirement paragraph	Test Method
Resistance to mustard (H)	3.3.3	4.4.2
Resistance to GB	3.3.3	4.4.3

4.4 Methods of inspection.

4.4.1 Weathering procedure. The test shall be performed with filters for a period of 100 hours on specimens 4 inches by 6 inches minimum with the heavily coated side of the cloth exposed as specified in Method 5804 of FED- STD- 191. After exposure, the specimens shall be allowed to dry and then be conditioned at standard conditions for 4 hours.

4.4.2 Resistance to mustard (H) test. Resistance to permeation by mustard (H) shall be determined as specified in Method 204.1.3 or 209.1.2 of MIL- STD- 282. Test specimens shall be obtained by quartering a 3- yard piece of the coated cloth and selecting one specimen from each quarter, with one determination made on the heavily coated side of each specimen. No test specimen shall be taken within 2 inches from the selvage edge. All specimens shall meet the

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requirements listed in table II. In the event that three of the four tested specimens meet the requirements but one does not, then two additional specimens from each quarter shall be tested with one determination made on each specimen. If seven of the eight specimens tested meet the requirements in table II and the eighth specimen is not less than 75 minutes, the material shall be considered acceptable.

4.4.3 Resistance to GB test. Resistance to permeation by GB shall be determined as specified in Method 206.1.5 or 208.1.2 of MIL- STD- 282. Test specimens shall be obtained by quartering a 3 yard piece or the coated cloth and selecting two samples, at random, from each quarter, with eight determinations made on the heavily coated side of the cloth. No test specimen shall be taken within 2 inches of the selvage edge. All samples shall meet the requirements listed in table II. In the event that seven or the tested specimens meet the requirements but one does not, four additional test specimens from each quarter shall be tested. If all sixteen specimens meet the requirements, the material shall be considered acceptable.

4.4.4 Resistance to mustard (H) after accelerated weathering. Test specimens shall be exposed to accelerated weathering as specified in 4.4.1 and tested for resistance to mustard (H) as specified in 4.4.2.

4.4.5 Resistance to GB after accelerated weathering. Test specimens shall first be exposed to accelerated weathering as specified in 4.4.1 and then tested for resistance to GB as specified in 4.4.3.

4.4.6 Resistance to mustard (H) after decontamination. Test specimens shall first be subjected to decontamination as specified in 4.4.9 and then tested for resistance to mustard (H) as specified in 4.4.2.

4.4.7 Resistance to GB after decontamination. Test specimens shall first be subjected to decontamination as specified in 4.4.9 and then tested for resistance to GB as specified in 4.4.3.

4.4.8 Colorfastness to decontamination. Test specimens shall be exposed to decontamination as specified in 4.4.9. The exposed specimen shall not show an “appreciable change in color” when compared to the unexposed material. “Appreciable change in color” means a change that is immediately noticeable in comparing the test specimen with the original unexposed material. If closer inspection or a change of angle of light is required to make apparent a slight change in color, the change is not considered appreciable.

4.4.9 Decontamination procedure. Strips of coated cloth, 10 inches by 22 inches, shall be cut along both the warp and filling directions. Immerse strips for 10 minutes in a decontaminating slurry made up of Super Tropical Bleach (STB) conforming to MIL- DTL- 12468 at a concentration of 50 pounds of STB per 5 gallons of water. (Note: STB is corrosive to most metals and is injurious to most fabrics. A protective mask and gloves should be worn, when handling this material.) At the end of 10 minutes, remove strips from the slurry and immerse in 4 gallons of

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boiling water for 2 hours. At the end of the 2 hours, remove strips from wash and air dry at room temperature.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in- house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The coated cloth is intended for use in the fabrication of impermeable clothing affording protection against toxicological warfare agents.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(c) Packaging requirements (see 5.1).

6.3 Shelf life. When a first article is required, it should be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Conductance of tests. Because of the extreme dangers involved in handling toxicological warfare agents, and the special equipment required in connection with their use, all document

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tests involving toxicological agents will be directed to the:

Edgewood Chemical Biological Center
ATTN: RDCB- DET- P
8820 Fleming Road
Building E5100 Room 16
APG, MD 21010- 5424

6.5 Suggested methods of coating cloth. In order to meet the requirements for toxicological agents resistance, a dense nonporous coating showing no voids at a cross section magnification of 60X has been found necessary. The following methods of coating cloth have been found to produce a coated cloth that will pass the toxicological agents resistance test:

a. A spreader operation in which not less than nine passes are made on each side of the base cloth, with drying passes after every third spreader pass.

b. A combination of calender and spread coating employing the use of a high pressure curing process.

6.6 Subject term (key word) listing.

Protection, chemical agent

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6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - EA
Navy – NU
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Preparing activity:
Army - EA
(Project 8305- 2018- 007)

Review activities:
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